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SUBJECT: Submission of Research Article: "Autonomous Multi-Physics Self-Regulation and Thermal-Stress Bounding in High-Temperature Solid-Oxide CO_2 Electrolysis and Sabatier Synthesis Loops" (Manuscript ID: APEN-D-26-26325)

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript titled **"Autonomous Multi-Physics Self-Regulation and Thermal-Stress Bounding in High-Temperature Solid-Oxide CO_2 Electrolysis and Sabatier Synthesis Loops"** for consideration as a Full Research Paper in *Applied Energy*. We explicitly select the **traditional no-fee subscription publishing model**.

Decarbonization of heavy industry and closed-loop carbon recycling rely heavily on high-enthalpy solid-oxide CO_2 electrolysis coupled with exothermic Sabatier hydrocarbon synthesis. However, scaling these systems to industrial continuous-flow baselines presents severe operational bottlenecks: microsecond kinetic stiffness, localized thermal runaway, and catastrophic thermo-mechanical stress cracking driven by thermal expansion mismatches.

In this work, we present **Project REGENESIS**, introducing a fundamental paradigm shift: **thermodynamic self-regulation**. We prove that cross-domain physical feedbacks—specifically mechanoelectrochemical d-band catalyst shifts ($\partial \eta_{act}/\partial \sigma_h = 0.04 V/MPa$), electro-thermal Ohmic dissipation ($\partial \dot{Q}/\partial \eta_{act} = 2.10 KV^{-1}s^{-1}$), and thermo-mechanical stress coupling ($\partial \sigma_h/\partial T = -3.50 MPa/K$)—enable closed-loop systems to achieve autonomous thermal-stress bounding and global spectral stability ($\mathrm{Re}(\lambda_{\max}) = -0.1389 < 0$) without requiring active external thermal control loops.

Key Quantitative Breakthroughs Presented in This Work:

1. **Continuous Scale Throughput:** Industrial scale-up validation across a 1,000 kg/hr continuous-flow operational baseline yielding a 94.8% methane conversion efficiency.
2. **Dynamic Shock Damping:** Operational transient simulations confirming $> 99.8\%$ exponential attenuation of severe combined thermal and electrical shocks ($\Delta T = -45K$, $\Delta V = +0.35V$) within a 60-second window.
3. **Out-of-Cluster Generalization:** Leave-One-Cluster-Out Cross-Validation (LOCO-CV) across five distinct operational regimes demonstrating 100% spectral stability ($\mathrm{Re}(\lambda_{\max}) \leq -0.1044$) and low generalization error (RMSE $\leq 3.47\%$).
4. **Structural Yield Bounding:** 3D anisotropic Tsai-Wu tensor failure analysis proving structural survivability ($FI = 0.682 \leq 0.75$, $+31.8\%$ safety reserve) under Pareto-optimized setpoints ($T_{op} = 1045.2K$, $\eta_{act} = 0.142V$).

5. **Industrial Durability:** Accelerated degradation modeling across 10,000 continuous operational hours demonstrating performance loss bounded at $< 1.85\%$ per 1,000 hours ($> 81.5\%$ total capacity retention).

Given *Applied Energy*'s focus on novel energy conversion systems, clean fuel synthesis, and industrial decarbonization, we believe this manuscript will resonate strongly with your readership. We confirm that this manuscript is original, has not been published previously, and is not currently under consideration elsewhere.

Thank you for your time and consideration.

Sincerely,

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